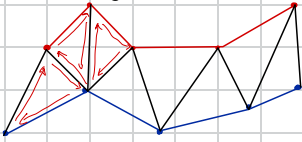


TD Meshing

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Exercise 1:

We need to triangulate two curve:



Condition : the two curve need to have the same number of vertices
: the two curve can't cross each other

Curve format : For each curve, we have a : Vertices[m] with m the number of point
tab1 & tab2

Pseudo code : triangulate

Triangulate() {

for (int i=0; i < tab1.length(); i++)

{

Halfedges he0, he1, he2;

he0->source = tab1[i];

he2->source = tab2[i];

he1->source = tab2[i+1];

he0->next = he1; he1->next = he2; he2->next = he0;

he0->prev = he2; he1->prev = he0; he2->prev = he1;

he0->adjacent_face = newFace //New face for the next faces)

newFace->adjacent_face = he0

if (twinT != null) {

he0->twin = twinT;

twinT->twin = he0;

}

twinT = he1;

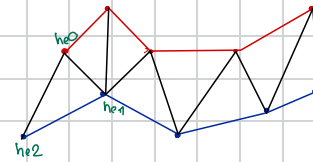
}

Add he0, he1, he2 to halfedge, newFace to faces;

i=0 : trace : (tab1[i], tab2[i])

i>1 : trace : (tab1[i], tab2[i-1])

trace : (tab1[i], tab2[i])



ReadFile : (file to convert the csv data input to usable data for the function).

```
void ReadFile ( csv file )  
{  
    int length = file.length / 2;  
    Vertices tab1[length], tab2[length];  
    foreach (line in file) {  
        fill tab1 & tab2 with data;  
    }  
}
```

format: tab, id, x, y

example: tab1, 0, 6, 2 → tab1[0] = (6, 2)

Exercice 2:

Condition: One base
Nb of curve ≥ 2